

TrES-3b

R-2781

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160720 · created 2026-09-04T00:00:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457589.8305 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.8 +/- 2.21 [%]
Orbital Inclination (inc)	83.87 +/- 2.86
Airmass coefficient 1 (a1)	0.66 +/- 0.15
Airmass coefficient 2 (a2)	0.31 +/- 0.17
Scatter in the residuals of the lightcurve fit is	21.0 %
Best Comparison Star	#3 - [452, 332]
Optimal Aperture	2.33
Optimal Annulus	10.3
Transit Duration (day)	0.066 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.195 $\pm$ 0.057 vs 0.16309 (deviation 0.3 σ)
Duration fit vs published	0.066 d vs 0.0567 d (deviation 0.24 σ)
Mid-time O-C	-3.8 min
Depth SNR	1.8
Residual scatter	21.0 %

Light curve

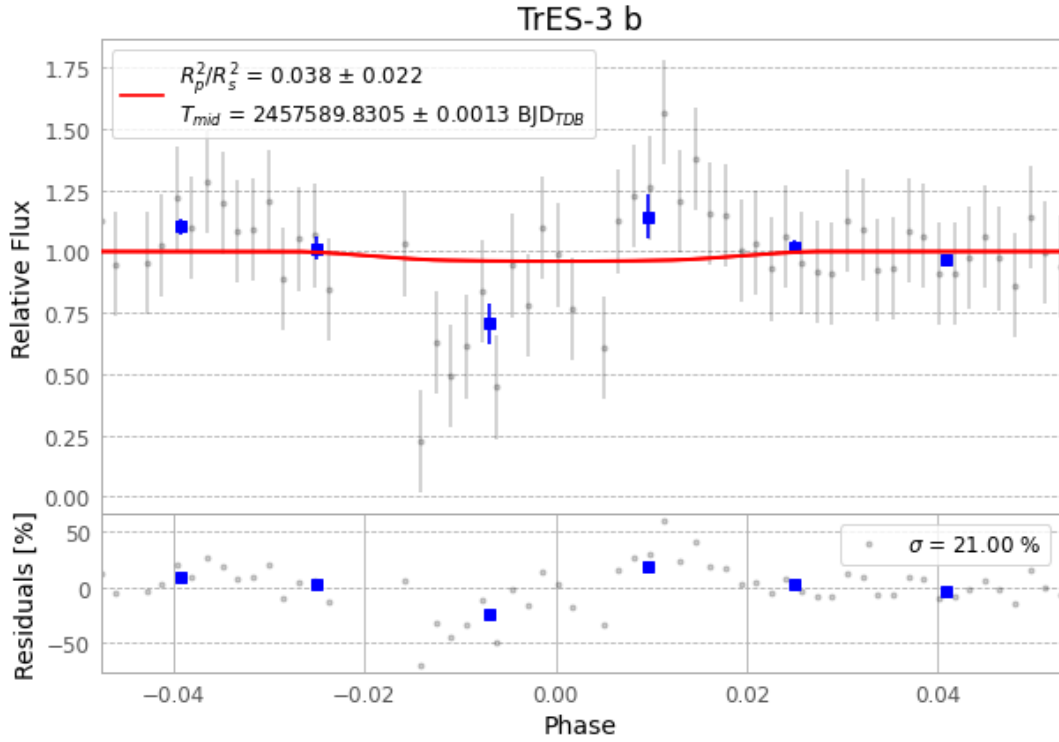


Figure 1 — FinalLightCurve_TrES-3 b_2016-07-19.png (SHA-256 026bc3e68ed23691...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [191, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 19afcfac2b3c5c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3160720062114.FITS → TRES-3160720093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160720.log (a588a05ebe63...), FinalLightCurve_TrES-3 b_2016-07-19.png (026bc3e68ed2...), FinalLightCurve_TrES-3 b_2016-07-19.csv (2531a221d519...)

Compute resources

Wall time 196.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:00Z.